Credal Uncertainty Quantification for Reliable Stance Detection on Unseen Targets

Abstract

Stance detection identifies whether the author of a text is in favor of, against, or neutral towards a given target. Current models output a single probability distribution over stance labels. That one distribution mixes up aleatoric uncertainty (the text is genuinely ambiguous and humans disagree about it) with epistemic uncertainty (the model has never seen this target or domain). As a result, stance detectors are over-confident on unseen targets and cannot give a principled reason to abstain. This project proposes a stance detector that outputs a credal set, i.e. a convex set of probability distributions, summarised as a lower and an upper probability for each stance. The credal set is built from a diverse ensemble of transformer and LLM backbones, decomposed into aleatoric and epistemic parts using upper and lower entropy, calibrated with conformal prediction to obtain a coverage guarantee, and turned into a decision (predict, return a set of plausible stances, or defer to a human) with imprecise-probability decision rules. No new annotation is collected: human label distributions come from the publicly released raw annotator labels of SemEval-2016. The method is evaluated in in-target, cross-target and zero-shot settings against seven families of uncertainty baselines. To our knowledge, this is the first application of credal uncertainty quantification to stance detection.

Contents

1 Introduction and Motivation **3**

2 Problem Statement **3**

3 Background **4**
3.1 Credal sets 4
3.2 Uncertainty measures on credal sets 5
3.3 Classical versus modern credal models 5

4 Related Work and Research Gap **5**

5 Proposed Solution: System Architecture **6**
5.1 M1: Data layer 6
5.2 M2: Human label distributions from existing raw annotations 7
5.3 M3: Input encoding 7
5.4 M4: Backbone ensembles 8
5.5 M5: Credal layer 8
5.6 M6: Uncertainty decomposition 8
5.7 M7: Conformal credal calibration 8
5.8 M8: Decision and routing 9
5.9 M9 (optional): concept-level credal network 9

6 Worked Examples **10**

7 Evaluation Plan **10**
7.1 Baselines 10
7.2 Metrics 10
7.3 Experiments 11
7.4 Ablations 12
7.5 Statistical rigour 12

8 Expected Outcomes **12**

9 Implementation and Timeline **12**

10 Risks and Mitigations **12**

11 Ethical Considerations **12**

12 Expected Contributions **13**

1 Introduction and Motivation

Stance detection takes a piece of text and a *target* (a person, policy, product or claim) and decides whether the author is FAVOR, AGAINST or NONE towards that target [1, 3, 4]. It is used in misinformation analysis, content moderation, political opinion tracking and argument mining.

A stance detector is normally a classifier that returns one probability vector, for example $p = [0.44, 0.41, 0.15]$ over $\{\text{FAVOR}, \text{AGAINST}, \text{NONE}\}$. That one vector cannot distinguish two very different situations:

1. **Ambiguity (aleatoric uncertainty).** The text itself is unclear. Example: *"Oh great, another 'climate emergency' summit. I'm sure the private jets will fix it."* Is the author for or against climate action? Even human annotators split on this. The VAST benchmark reports a Krippendorff's α of only 0.427 [5]. No amount of extra training data will remove this uncertainty, because it lives in the text.

2. **Ignorance (epistemic uncertainty).** The model lacks knowledge. Example: a tweet about the target *"Digital euro"* when the model was trained on tweets about abortion, feminism and atheism. More data about the new target *would* remove this uncertainty.

Both situations can produce the same output vector. In the first case the right action is to flag the text as contested; in the second it is to defer to a human or to collect labels for the new target. A single probability vector gives no basis for choosing between them. Recent analyses of LLM stance detection reach the same conclusion: hard instances are usually ambiguous or context-dependent, and systems used in sensitive settings should report uncertainty and allow human review [8].

This project addresses the gap with **credal sets**: instead of one distribution, the model predicts a *set* of plausible distributions. The size and shape of that set separate ambiguity from ignorance, and imprecise-probability decision theory tells us what to do with it.

2 Problem Statement

Given a text x and a target t, existing stance detectors output a single probability vector $p(y \mid x, t)$ over $\mathcal{Y} = \{\text{FAVOR}, \text{AGAINST}, \text{NONE}\}$. This vector conflates (i) *aleatoric* uncertainty, arising from sarcasm, implicit stance, missing context and genuine human disagreement, with (ii) *epistemic* uncertainty, arising from unseen targets, domain shift and limited data. Consequently, models are over-confident on unseen targets and cannot justify abstention.

We propose a stance detector that outputs a *credal set* $\mathcal{C}(x, t) \subseteq \Delta_K$ of distributions, represented by class-wise lower and upper probabilities $[\underline{P}(y), \overline{P}(y)]$, built from ensembles of transformer and LLM backbones. The system (a) separates aleatoric from epistemic uncertainty, (b) provides a distribution-free conformal coverage guarantee for the human label distribution, and (c) uses the credal set to predict, return a set of plausible stances, or defer to a human. It is evaluated on in-target, cross-target and zero-shot benchmarks, and against human label distributions derived from existing raw multi-annotator stance data.

Research questions.

RQ1 Do credal stance models give better uncertainty estimates than standard methods (softmax, MC Dropout, deep ensembles, evidential deep learning), in-target and across targets?

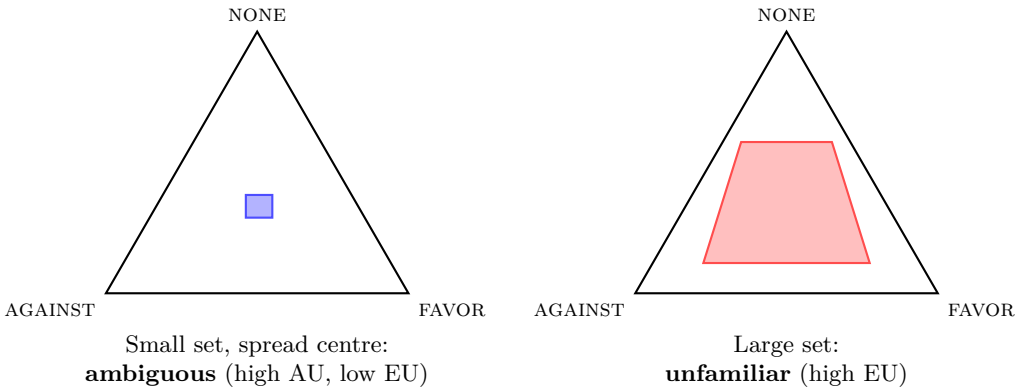


Figure 1: Two credal sets on the stance simplex. Each point in a triangle is a probability vector over the three stances. Left: the model is confident that the distribution is spread between FAVOR and AGAINST, i.e. the text is contested. Right: the model cannot narrow down the distribution at all, i.e. it has not learned this target.

RQ2 Does the credal decomposition separate ambiguity from ignorance? Does the epistemic part rise on unseen targets, and does the aleatoric part track human disagreement?

RQ3 Can we guarantee, with high probability, that the predicted set covers the true human label distribution?

RQ4 Does credal uncertainty lead to better decisions (abstention, deferral, set-valued prediction) than confidence thresholding?

RQ5 Do credal sets built from LLM ensembles beat verbalised and sampling-based LLM confidence?

Scope. The claim is about *reliability*, not about a new state-of-the-art macro-F1. The system classifies texts, never the beliefs of identifiable people.

3 Background

3.1 Credal sets

Let $K = 3$ and let Δ_K be the probability simplex. A **credal set** is a closed convex set $\mathcal{C} \subseteq \Delta_K$ of candidate distributions for an imprecisely known ground truth [9]. We use *probability-interval* credal sets, defined by lower bounds l and upper bounds u per class:

$$\mathcal{C}(l, u) = \{\, p \in \Delta_K : l_k \le p_k \le u_k \text{ for all } k \,\}. \tag{1}$$

The lower and upper probabilities are $\underline{P}(k) = \min_{p \in \mathcal{C}} p_k$ and $\overline{P}(k) = \max_{p \in \mathcal{C}} p_k$.

Figure 1 shows why a credal set carries two kinds of information. A *small* set centred on a spread-out distribution means the model is sure *that* the text is ambiguous (aleatoric). A *large* set means the model does not know which distribution is right (epistemic).

	Classical credal models	Modern credal deep learning
Examples	Credal networks [11], Naive Credal Classifier [12]	Credal Wrapper [18], Credal Deep Ensembles [19], CreINNs [20], conformal credal sets [21], credal LLMs [23]
Where the credal set lives	Inside the model (imprecise conditional probability tables)	At the output (lower/upper class probabilities)
Input data	Discrete, tabular	Any: images, text, graphs, LLM tokens
Why	Exact inference needs few discrete variables	The encoder handles the raw input

Table 1: The two meanings of "credal" in machine learning. This project uses the modern kind; a classical credal network appears only as an optional interpretability module (Section 5.9).

3.2 Uncertainty measures on credal sets

Following Abellán et al. [13] and Hüllermeier and Waegeman [14], with H the Shannon entropy in bits:

$$\mathrm{TU}(\mathcal{C}) = \overline{H}(\mathcal{C}) = \max_{p \in \mathcal{C}} H(p) \quad \text{(total uncertainty)}, \tag{2}$$

$$\mathrm{AU}(\mathcal{C}) = \underline{H}(\mathcal{C}) = \min_{p \in \mathcal{C}} H(p) \quad \text{(aleatoric)}, \tag{3}$$

$$\mathrm{EU}(\mathcal{C}) = \overline{H}(\mathcal{C}) - \underline{H}(\mathcal{C}) \quad \text{(epistemic)}. \tag{4}$$

For $K = 3$ the maximum entropy is $\log_2 3 \approx 1.585$ bits. Sale et al. [15] show that the *volume* of a credal set is a sound epistemic measure only in the binary case, which is why we use the entropy decomposition for three classes.

3.3 Classical versus modern credal models

A frequent objection is that "credal networks only work on tabular data". This is true of *classical* credal networks [11] and the Naive Credal Classifier [12]: they place imprecise probability tables *inside* a graphical model over discrete variables, and exact inference is intractable for more than a handful of such variables. One cannot build a credal network over a 30,000-word vocabulary.

Modern *credal deep learning* is different (Table 1). The credal set lives at the *output* of a neural network, as lower and upper class probabilities. The raw input, whether image, text or graph, is handled by an ordinary encoder; the credal construction only ever sees probability vectors. Published evidence on language data includes credal sets on the ChaosNLI natural-language-inference benchmark [21, 22], credal LLMs built from LoRA ensembles [23], and a credal attention mechanism inside the Transformer itself [24]. Stance detection has the same structure as NLI (a text pair and three classes), and EZ-Stance even reformulates zero-shot stance detection *as* NLI [6]. The machinery therefore carries over directly.

4 Related Work and Research Gap

Stance detection. Benchmarks include SemEval-2016 [1] (five targets, tweets), VAST [5] (thousands of zero-shot topics) and EZ-Stance [6] (47k pairs, target- and domain-based zero-shot subtasks). LLM stance detection suffers from sentiment–stance spurious correlations and target bias; FACTUAL [7] corrects this with a calibration network, but still outputs a single distribution.

Uncertainty in NLP. Deep ensembles [33], MC Dropout [34], evidential deep learning [36] and conformal prediction [37] are the standard tools. Human label variation is increasingly treated as signal rather than noise [27, 28]; ChaosNLI collects about 100 annotations per NLI item [26], and multi-perspective stance detection trains on individual annotator labels [29].

Credal deep learning. The Credal Wrapper [18] builds a credal set from any ensemble post hoc; Credal Deep Ensembles [19] and CreINNs [20] train networks to output intervals directly; conformalized credal set predictors [21] add coverage guarantees; EffCre [22] obtains credal sets from a single model; credal LLMs [23] use LoRA ensembles. Decision making with credal sets uses interval dominance and related rules [16], and credal classifiers are evaluated with utility-discounted accuracy [17].

Gap. A literature search (October 2026) found no published work that applies credal sets or imprecise probabilities to stance detection. The closest work is on NLI. The combination of (i) credal uncertainty decomposition, (ii) conformal guarantees for the human label distribution and (iii) imprecise-probability decision rules has not been studied for stance.

5 Proposed Solution: System Architecture

Figure **??** shows the end-to-end pipeline. Text and target enter a diverse ensemble; the ensemble's M probability vectors are turned into a credal set; the credal set is decomposed into aleatoric and epistemic uncertainty, calibrated with conformal prediction, and converted into a decision.

5.1 M1: Data layer

Table 2 lists the benchmarks and their roles. All sources are mapped to a unified label set $\{\text{FAVOR} = 0, \text{AGAINST} = 1, \text{NONE} = 2\}$.

Dataset	Domain	Targets	Role
SemEval-2016 Task 6 [1]	Tweets	5 fixed targets	In-target, leave-one-target-out
VAST [5]	NYT comments	∼5,600 topics	Zero-shot
EZ-Stance [6]	Tweets	Noun-phrase + claim targets	Zero-shot (target- and domain-based)
ChaosNLI [26]	NLI	–	Secondary conformal check (100 annotators/item)

Table 2: Datasets. Sizes should be verified against the official releases; tweet datasets are distributed as IDs and some tweets are no longer retrievable.

Evaluation splits.

- **In-target**: official SemEval train/test split.
- **Leave-one-target-out (LOTO)**: train on four SemEval targets, test on the fifth; five folds. A controlled epistemic shift.
- **Zero-shot**: VAST and EZ-Stance official zero-shot test sets. A strong epistemic shift.
- **Seen/unseen pool**: in-target test items (label 0) mixed with zero-shot test items (label 1), for unseen-target detection.

5.2 M2: Human label distributions from existing raw annotations

The project collects **no new annotations**. SemEval-2016 publicly releases every crowd worker's individual stance label for each tweet [1, 2]. We inspected the released file (October 2026); its contents are summarised in Table 3.

Subset	Count	Note
Shared-task tweets with all votes	4,306	median 8 annotators per tweet
Low-agreement tweets (dropped from the task, agreement $< 60\%$)	1,385	mean label entropy 1.445 bits (max 1.585)
Targets covered	5	Atheism, Feminist Movement, Hillary Clinton, Legalization of Abortion, Donald Trump

Table 3: Contents of the SemEval-2016 raw stance annotation file.

For each tweet x with n votes we compute the empirical human label distribution and its entropy:

$$\lambda_x = \frac{(n_{\text{FAVOR}},\, n_{\text{AGAINST}},\, n_{\text{NONE}})}{n}, \qquad H(\lambda_x) = -\sum_k \lambda_{x,k} \log_2 \lambda_{x,k}. \tag{5}$$

The raw NEUTRAL answer is mapped to NONE, matching the official label derivation. Because $n \approx 8$ is small, we also report a Dirichlet-smoothed version $(n_k + 0.5)/(n + 1.5)$ and repeat the conformal analysis on the $n \geq 10$ subset.

From this file we build five human-grounded splits:

HLV-train official SemEval train tweets minus HLV-cal (member training).

HLV-cal 20% of the train tweets, stratified by target and held out from training (conformal calibration).

HLV-test official SemEval test tweets with raw votes (human-alignment and coverage evaluation).

HLV-unseen
every tweet about one target (e.g. *Donald Trump*) that is excluded from training (human-labelled unseen-target test).

HLV-ambig
the 1,385 low-agreement tweets (ambiguity stress test; AU should be high here, and EU should *not* be driven by it).

5.3 M3: Input encoding

Encoder path. Sentence-pair input with the target first so it is never truncated:

```
[CLS] Climate Change is a Real Concern [SEP] We need to act NOW.
#ClimateAction [SEP]
```

LLM path. An instruction prompt (*"Target: ... Text: ... What is the author's stance towards the target? Answer with exactly one word: favor, against, or none."*) followed by a *verbalizer readout*: the next-token logits of the three answer words are passed through a softmax restricted to those three tokens. This yields a proper 3-class probability vector with no free-text generation and no parsing errors.

5.4 M4: Backbone ensembles

The credal set is only as informative as the ensemble is diverse, so diversity is engineered explicitly.

	Path A: encoder ensemble	Path B: LLM LoRA ensemble
Backbone	DeBERTa-v3-large [30]	Qwen2.5-7B-Instruct (frozen)
Members	$M = 8$, full fine-tuning	$M = 8$ LoRA adapters [31, 32]
Diversity	Different seeds; 80% bootstrap of training data per member	Seeds; 80% bootstrap; two instruction paraphrases
Loss	Cross-entropy on hard labels	Cross-entropy on the 3-token restricted softmax
Optimiser	AdamW, lr 10^{-5}, 6 epochs, early stopping	AdamW, lr 10^{-4}, cosine, 2 epochs

Table 4: The two backbone paths. Member logits are cached to disk once, so all credal, conformal and decision experiments run on CPU in seconds.

The output of M4 for an input (x, t) is a matrix $P \in \mathbb{R}^{M \times K}$ whose row m is $p_m = \text{softmax}(f_m(x, t))$.

5.5 M5: Credal layer

Construction (Credal Wrapper [18]). From the M member distributions take the class-wise extremes:

$$l_k = \min_m p_{m,k}, \qquad u_k = \max_m p_{m,k}, \qquad \mathcal{C}(x) = \{p \in \Delta_K : l \le p \le u\}. \tag{6}$$

Each bound is attained by some member and every member lies in $\mathcal{C}(x)$, so the bounds are reachable and no tightening step is needed.

Point prediction (intersection probability [18]). For metrics that need one distribution:

$$\alpha = \frac{1 - \sum_k l_k}{\sum_k (u_k - l_k)}, \qquad p_k^* = l_k + \alpha\,(u_k - l_k), \qquad \hat{y} = \arg\max_k p_k^*. \tag{7}$$

Alternatives compared in ablation. Credal sets from MC-Dropout samples; Credal Deep Ensembles, which train each network to output intervals directly [19]; CreINNs [20]; EffCre, which needs only a single model [22]; and credal ensemble distillation into one student [25].

5.6 M6: Uncertainty decomposition

Upper entropy $\overline{H}$ is a concave maximisation over a box-constrained simplex. From the KKT conditions the maximiser is $p_k = \text{clip}(\tau, l_k, u_k)$ for a scalar τ found by bisection so that $\sum_k p_k = 1$ ("water-filling").

Lower entropy $\underline{H}$: entropy is concave, so its minimum over the polytope is attained at a vertex. For $K = 3$ there are at most 12 candidate vertices, which are enumerated directly.

Epistemic uncertainty is then $\text{EU} = \overline{H} - \underline{H}$. Table 5 gives the interpretation.

5.7 M7: Conformal credal calibration

Ensemble credal sets carry no formal guarantee. Following Javanmardi et al. [21], we calibrate on HLV-cal, where the target is the full human label distribution λ_x rather than a single label:

1. Predictor: $g(x) = p^*(x)$ from Eq. (7).

2. Non-conformity score on each calibration item: $s_i = \text{TV}(\lambda_i, g(x_i)) = \frac{1}{2} \sum_k |\lambda_{i,k} - g_k(x_i)|$.

AU	EU	Meaning	Typical cause
low	low	Confident, clear	Explicit stance on a seen target
high	low	Ambiguous / contested	Sarcasm, mixed stance; humans also disagree
any	high	Unfamiliar	Unseen target, new domain, emerging event
high	high	Both	Ambiguous text on an unseen target

Table 5: Reading the decomposition.

3. Quantile: with n calibration items and miscoverage α, $\hat{q}$ is the $\lceil (n+1)(1-\alpha) \rceil$-th smallest score.

4. Conformal credal set: $\hat{\mathcal{C}}(x) = \{\lambda \in \Delta_K : \mathrm{TV}(\lambda, g(x)) \leq \hat{q}\}$.

5. Guarantee (exchangeability [37]): $\Pr\left(\lambda_x \in \hat{\mathcal{C}}(x)\right) \geq 1 - \alpha$.

The class-wise bounds of this TV ball have closed form, $\hat{l}_k = \max(0, g_k - \hat{q})$ and $\hat{u}_k = \min(1, g_k + \hat{q})$, and feed the decision layer. The ensemble supplies the *shape* (through g) and conformal prediction supplies a *guaranteed size* (through $\hat{q}$).

5.8 M8: Decision and routing

Interval dominance [16]. Class j dominates class k if $l_j > u_k$, i.e. j is more probable than k under *every* distribution in the credal set. The non-dominated set is

$$\mathrm{ND}(x) = \{\, k : u_k \geq \max_j l_j \,\}. \tag{8}$$

Figure 5.8 shows the routing policy. Thresholds τ_{AU} and τ_{EU} are chosen on the development set to meet a deferral budget (e.g. at most 10% of traffic deferred).

Routing policy. A single non-dominated class is predicted; a contested text is predicted but flagged; several non-dominated classes with low epistemic uncertainty give a set-valued answer; high epistemic uncertainty defers to a human.

A cost-sensitive variant assigns $c_{\mathrm{err}} = 1$ to a wrong single label, $c_{\mathrm{defer}} = 0.3$ to a deferral and $c_{\mathrm{set}} = 0.15\,(|\mathrm{ND}| - 1)$ to a set-valued answer, and reports expected cost against deferral rate.

5.9 M9 (optional): concept-level credal network

For interpretability, an LLM extracts a handful of discrete concepts from the text (*target mention*: explicit/implicit/absent; *sentiment*: pos/neg/neutral; *sarcasm*: yes/no; *negation*: yes/no). A

classical credal network [11] over these concepts, with imprecise conditional tables learnt via the Imprecise Dirichlet Model [10], then gives lower and upper $P(\text{stance} \mid \text{concepts})$. This is where a classical credal network fits: on a small table of discrete features, not on raw text. It explains *why* a prediction is uncertain (e.g. "sarcasm = yes and this configuration is rare in training").

6 Worked Examples

Table 6 traces three inputs through M5–M8 with $M = 5$ members. All numbers were produced by the reference implementation of the credal core.

	Bounds $[l_k, u_k]$					
Example	FAVOR	AGAINST	NONE	AU	EU	Decision
A. Clear, seen target	$[0.88, 0.92]$	$[0.04, 0.07]$	$[0.04, 0.05]$	0.48	0.17	Predict FAVOR
B. Sarcasm, seen target	$[0.42, 0.45]$	$[0.40, 0.43]$	$[0.14, 0.15]$	1.44	0.02	Set-valued {FAVOR, AGAINST}, flagged
C. Unseen target	$[0.15, 0.80]$	$[0.10, 0.75]$	$[0.10, 0.50]$	0.92	0.66	Defer

Table 6: Three worked examples (entropies in bits). Example A: all members agree on a confident FAVOR. Example B: all members agree that the text is split between FAVOR and AGAINST; the set is small (low EU) but centred on a spread distribution (high AU). Example C: the members disagree wildly; the set is large (high EU).

The key observation is Example C. Averaging the five members gives $[0.43, 0.38, 0.19]$, which looks like a mild FAVOR and hides the disagreement entirely. The credal set exposes it: $\overline{P}(\text{FAVOR}) = 0.80$ but $\underline{P}(\text{FAVOR}) = 0.15$. Conversely, in Example B a softmax model would output FAVOR with $p = 0.44$ and give no sign that the text is contested, whereas the credal set shows a small, spread-out set. In the SemEval raw annotations, such texts are exactly where human votes split (e.g. 3 FAVOR, 4 AGAINST, 1 NONE; $H = 1.41$ bits).

Conformal example. If $g(x) = [0.70, 0.20, 0.10]$ and the calibrated radius is $\hat{q} = 0.15$, the calibrated bounds are $\hat{l} = [0.55, 0.05, 0.00]$ and $\hat{u} = [0.85, 0.35, 0.25]$. FAVOR dominates ($0.55 > 0.35$), so the system predicts FAVOR, and with probability at least 0.9 the true human distribution lies within TV distance 0.15 of $g(x)$.

7 Evaluation Plan

7.1 Baselines

7.2 Metrics

Predictive quality
F_{avg} (official SemEval metric), macro-F1, accuracy.

Calibration
expected calibration error (15 bins), negative log-likelihood, Brier.

Uncertainty quality
misclassification AUROC/AUPR using TU; **unseen-target AUROC** using EU; area under the risk–coverage curve [39].

Set-valued prediction
determinacy, single accuracy, set accuracy, and utility-discounted accuracies u_{65} and u_{80} [17].

ID	Baseline	Note
B1	Single model, softmax confidence	max-probability and entropy
B2	Temperature scaling [35]	fitted on dev
B3	MC Dropout [34]	$M = 8$ samples; entropy and mutual information
B4	Deep ensemble, mean prediction [33]	**same members as the credal model**
B5	Evidential deep learning [36]	Dirichlet head; vacuity as EU
B6	Label-level conformal (LAC, APS) [37, 38]	set size and coverage
B7	LoRA ensemble, mean prediction [32]	LLM equivalent of B4
B8	LLM verbalised / sampling confidence [40]	zero-shot
B9	Prompt-credal (24 prompt variants, no fine-tuning)	ours; tests prompt diversity alone
B10	FACTUAL [7]	stance-specific LLM calibration
B11	Multi-perspective soft-label training [29]	human-label-variation baseline

Table 7: Baselines. B4 uses exactly the same trained members as the credal model, so any difference comes from the credal representation and decision rule, not from extra compute.

Human alignment

Spearman $\rho(\mathrm{AU}, H(\lambda_x))$ (expected positive) versus $\rho(\mathrm{EU}, H(\lambda_x))$ (expected weaker); Jensen–Shannon divergence between p^* and λ_x; AU on HLV-ambig versus HLV-test.

Conformal validity

empirical coverage at $\alpha \in \{0.05, 0.1, 0.2\}$; mean $\hat{q}$ and mean |ND| (efficiency); coverage per human-entropy bin.

7.3 Experiments

Exp.	Question	Data	Primary metrics
E1	Does the credal model hurt in-target accuracy?	SemEval in-target	F_{avg}, ECE, AURC
E2	Cross-target reliability	SemEval LOTO	macro-F1, misclassification AUROC, AURC
E3	Zero-shot reliability	VAST, EZ-Stance	macro-F1, ECE, AURC, u_{65}
E4	Does EU detect unseen targets?	Seen/unseen pool; HLV-unseen	unseen-target AUROC
E5	Does AU match human disagreement?	HLV-test, HLV-unseen, HLV-ambig	$\rho(\mathrm{AU}, H(\lambda))$, JSD
E6	Conformal coverage	HLV-cal $\rightarrow$ HLV-test; ChaosNLI	coverage, efficiency
E7	Decision value	all test sets	expected cost vs. deferral rate
E8	LLM path	VAST, EZ-Stance, HLV-test	all of the above

Table 8: Experiment matrix.

7.4 Ablations

Ensemble size $M \in \{2, 4, 8, 16\}$; credal construction (wrapper, MC-Dropout, CreDE, CreINN, EffCre, distilled); diversity source (seed, bootstrap, prompt paraphrase); conformal score (TV, KL, Wasserstein); hard versus soft-label training; backbone (DeBERTa-v3, BERTweet, Qwen2.5-7B, Llama-3.1-8B); uncertainty measure (entropy decomposition versus width versus volume); and active learning on a new target by EU-sampling versus random.

7.5 Statistical rigour

Three independent ensemble pools with disjoint seeds (mean ± std); 95% bootstrap confidence intervals; paired permutation tests against B4 with Holm–Bonferroni correction; conformal results averaged over 100 random calibration/test re-splits.

8 Expected Outcomes

Hypothesis	Expected result
H1 (RQ1)	Higher misclassification AUROC and lower AURC than B1–B5 at equal macro-F1 (± 1 point).
H2a (RQ2)	EU is significantly higher on unseen than on seen targets; unseen-target AUROC ≥ 0.75.
H2b (RQ2)	$\rho(\mathrm{AU}, H(\lambda_x)) \geq 0.4$ and larger than $\rho(\mathrm{EU}, H(\lambda_x))$.
H3 (RQ3)	Conformal credal sets reach coverage $1 - \alpha$ ($\pm 2\%$) and are smaller when built on the credal score than on the softmax score.
H4 (RQ4)	Credal routing gives lower expected cost than confidence thresholding at every deferral budget.
H5 (RQ5)	LoRA-ensemble credal LLMs have lower ECE than verbalised and sampling-based LLM confidence.

Table 9: Hypotheses and the results that would confirm them.

9 Implementation and Timeline

Stack. PyTorch; Hugging Face `transformers` and `peft`; a small NumPy module for the credal core (bounds, intersection probability, upper/lower entropy, dominance, conformal quantile) with unit tests; Hydra for configuration; Gradio for the demonstration dashboard. Member logits are cached once, so every credal and decision experiment is a CPU job.

Compute. Roughly 1–3 GPU-hours per DeBERTa member per dataset and 3–8 GPU-hours per LoRA adapter on a single A100-80GB; three pools of eight members per path.

10 Risks and Mitigations

11 Ethical Considerations

The system classifies the stance of *texts* and must not be used to infer or score the beliefs of identifiable people; large-scale stance inference raises surveillance risks [8]. The defer and "contested" outputs are designed to *reduce* over-confident labelling of political speech. No new human data is collected; the SemEval annotator IDs are already anonymised and are not used to profile workers. Disagreement is treated as signal rather than noise [27, 28]. The SemEval

Weeks	Work	Milestone
1–2	Literature review (confirm novelty); download datasets incl. SemEval raw annotations; repository, configs	Literature document; data inventory
3–4	Data pipeline; λ_x and HLV splits; baselines B1–B3	Baseline table; HLV statistics
5–6	Train DeBERTa ensemble pools; cache logits	Cached logits (Path A)
7–8	Credal core and unit tests; E1–E3; baselines B4–B6	First credal vs. baseline results
9–10	E4 unseen-target detection; decision layer; E7 cost curves	Routing results
11–12	E5 human alignment; E6 conformal coverage (+ ChaosNLI)	Mid-term review: working system with coverage guarantee
13–15	LoRA ensembles on the LLM; B7–B10; E8	LLM credal results
16–18	Ablations; credal variants; three-pool variance runs; significance tests	Full results tables
19–20	Error analysis by human-entropy bin and implicit-stance slice; M9 if time; demo	Demo v1; qualitative analysis
21–23	Paper writing; figures; code clean-up	Paper draft
24–26	Review, revisions, submission; final report	Final submission and presentation

Table 10: Six-month plan.

Risk	Mitigation
Credal model gives no F1 gain	Expected; the claim is about reliability metrics (AURC, unseen-target AUROC, coverage), not F1.
Ensemble members too similar, so credal sets are narrow	Bootstrap + seed + prompt diversity; diversity ablation; Credal Deep Ensembles, which learn intervals directly.
Only ≈ 8 votes per tweet, so λ_x is noisy	Dirichlet smoothing; $n \geq 10$ subset; confirm coverage on ChaosNLI with ≈ 100 votes.
Raw votes exist only for SemEval targets	Human-alignment claims are scoped to SemEval; VAST and EZ-Stance use hard-label metrics only.
Conformal exchangeability violated across sources	Calibrate per source; report conditional coverage; state as a limitation.
Prior credal stance work appears	Literature check in weeks 1–2; the decomposition + conformal + decision combination remains novel.

Table 11: Main risks.

raw annotations may not be redistributed, so the repository ships a download script, and tweets are shared as IDs only.

12 Expected Contributions

1. The first credal uncertainty framework for stance detection, with an explicit aleatoric/epistemic decomposition and imprecise-probability decision making.

2. A human-label-distribution benchmark for stance built from the existing SemEval-2016 raw annotations (HLV-test, HLV-unseen and HLV-ambig splits), with released processing scripts.

3. Conformal credal guarantees for stance: coverage of the *human label distribution*, not just a single label.

4. An empirical study across in-target, cross-target and zero-shot settings and across encoder and LLM backbones, against eleven baselines.

5. Open-source code (cached-logit pipeline, tested credal core) and an interactive demo showing lower and upper stance probabilities.

References

[1] S. M. Mohammad, S. Kiritchenko, P. Sobhani, X. Zhu, and C. Cherry. SemEval-2016 Task 6: Detecting stance in tweets. In *Proceedings of SemEval*, 2016.

[2] S. M. Mohammad, P. Sobhani, and S. Kiritchenko. Stance and sentiment in tweets. *ACM Transactions on Internet Technology*, 17(3), 2017. Raw annotations: `https://saifmohammad.com/WebPages/StanceDataset.htm`.

[3] D. Küçük and F. Can. Stance detection: A survey. *ACM Computing Surveys*, 53(1), 2020.

[4] A survey of stance detection on social media: New directions and perspectives. *arXiv preprint arXiv:2409.15690*, 2024.

[5] E. Allaway and K. McKeown. Zero-shot stance detection: A dataset and model using generalized topic representations. In *Proceedings of EMNLP*, 2020.

[6] C. Zhao and C. Caragea. EZ-STANCE: A large dataset for English zero-shot stance detection. In *Proceedings of ACL*, 2024.

[7] A. Li et al. Mitigating biases of large language models in stance detection with counterfactual augmented calibration. *arXiv preprint arXiv:2402.14296*, 2024.

[8] SICI: A semantic-pragmatic complexity index reveals regime shifts in LLM stance detection. *arXiv preprint arXiv:2606.13189*, 2026.

[9] P. Walley. *Statistical Reasoning with Imprecise Probabilities*. Chapman & Hall, 1991.

[10] P. Walley. Inferences from multinomial data: Learning about a bag of marbles. *Journal of the Royal Statistical Society B*, 58(1):3–57, 1996.

[11] F. G. Cozman. Credal networks. *Artificial Intelligence*, 120(2):199–233, 2000.

[12] G. Corani and M. Zaffalon. Learning reliable classifiers from small or incomplete data sets: The naive credal classifier 2. *Journal of Machine Learning Research*, 9:581–621, 2008.

[13] J. Abellán, G. J. Klir, and S. Moral. Disaggregated total uncertainty measure for credal sets. *International Journal of General Systems*, 35(1):29–44, 2006.

[14] E. Hüllermeier and W. Waegeman. Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine Learning*, 110:457–506, 2021.

[15] Y. Sale, M. Caprio, and E. Hüllermeier. Is the volume of a credal set a good measure for epistemic uncertainty? In *Proceedings of UAI*, 2023.

[16] M. C. M. Troffaes. Decision making under uncertainty using imprecise probabilities. *International Journal of Approximate Reasoning*, 45(1):17–29, 2007.

[17] M. Zaffalon, G. Corani, and D. Mauá. Evaluating credal classifiers by utility-discounted predictive accuracy. *International Journal of Approximate Reasoning*, 53(8):1282–1301, 2012.

[18] K. Wang, F. Cuzzolin, et al. Credal wrapper of model averaging for uncertainty estimation in classification. In *Proceedings of ICLR*, 2025.

[19] K. Wang, F. Cuzzolin, S. Kudukkil Manchingal, K. Shariatmadar, D. Moens, and H. Hallez. Credal deep ensembles for uncertainty quantification. In *Advances in Neural Information Processing Systems*, 2024.

[20] K. Wang et al. CreINNs: Credal-set interval neural networks for uncertainty estimation in classification tasks. *Neural Networks*, 2025.

[21] A. Javanmardi, D. Stutz, and E. Hüllermeier. Conformalized credal set predictors. In *Advances in Neural Information Processing Systems*, 2024.

[22] P. Hofman, T. Löhr, M. Muschalik, Y. Sale, and E. Hüllermeier. Efficient credal prediction through decalibration. In *Proceedings of ICLR*, 2026.

[23] S. Kudukkil Manchingal, S. Nikolenko, and F. Cuzzolin. Credal large language models for semantic commitment under uncertainty. *arXiv preprint arXiv:2608.23244*, 2026.

[24] Credal Transformer: A principled approach for quantifying and mitigating hallucinations in large language models. *arXiv preprint arXiv:2510.12137*, 2025.

[25] Credal ensemble distillation for uncertainty quantification. *arXiv preprint arXiv:2511.13766*, 2025.

[26] Y. Nie, X. Zhou, and M. Bansal. What can we learn from collective human opinions on natural language inference data? In *Proceedings of EMNLP*, 2020.

[27] B. Plank. The "problem" of human label variation: On ground truth in data, modeling and evaluation. In *Proceedings of EMNLP*, 2022.

[28] A. Uma, T. Fornaciari, D. Hovy, S. Paun, B. Plank, and M. Poesio. Learning from disagreement: A survey. *Journal of Artificial Intelligence Research*, 72:1385–1470, 2021.

[29] B. Muscato, P. Bushipaka, G. Gezici, L. Passaro, and F. Giannotti. Multi-perspective stance detection. *arXiv preprint arXiv:2411.08752*, 2024.

[30] P. He, J. Gao, and W. Chen. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *Proceedings of ICLR*, 2023.

[31] E. J. Hu et al. LoRA: Low-rank adaptation of large language models. In *Proceedings of ICLR*, 2022.

[32] X. Wang, L. Aitchison, and M. Rudolph. LoRA ensembles for large language model fine-tuning. *arXiv preprint arXiv:2310.00035*, 2023.

[33] B. Lakshminarayanan, A. Pritzel, and C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, 2017.

[34] Y. Gal and Z. Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of ICML*, 2016.

[35] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *Proceedings of ICML*, 2017.

[36] M. Sensoy, L. Kaplan, and M. Kandemir. Evidential deep learning to quantify classification uncertainty. In *Advances in Neural Information Processing Systems*, 2018.

[37] A. N. Angelopoulos and S. Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4), 2023.

[38] Y. Romano, M. Sesia, and E. Candès. Classification with valid and adaptive coverage. In *Advances in Neural Information Processing Systems*, 2020.

[39] Y. Geifman and R. El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017.

[40] M. Xiong et al. Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs. In *Proceedings of ICLR*, 2024.